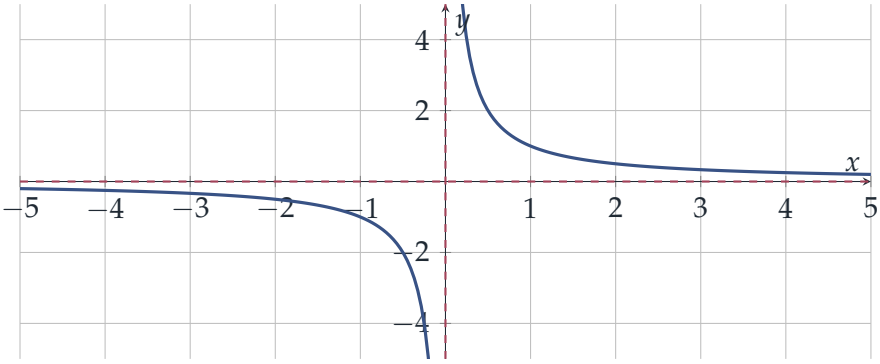
Chapter 4

Reciprocal function

Competences: determining a domain, drifting, studying a sign, drawing up a table of variations, representing

4.1 Reference course

The reciprocal function is defined by $f(x) = \frac{1}{x}$ on $\mathbb{R} \setminus \{0\}$.



Key ideas

$f'(x) = -\frac{1}{x^2} < 0$ for all $x \neq 0$. The reciprocal function is strictly decreasing on each intervals $]-\infty, 0[$ and $]0, +\infty[$. The rights $x = 0$ and $y = 0$ are asymptotes.

Frequent error

We cannot write that the reciprocal function is decreasing "on $\mathbb{R} \setminus \{0\}$ » without specifying the two intervals. $-1 < 1$ but $f(-1) = -1 < 1 = f(1)$, which would contradict the overall definition of decline.

For a function $g(x) = P(x) + \frac{k}{x}$, we proceed in this order : domain ; derivative ; put to the same denominator and factorization; derivative sign; variations; boundary or boundary behavior; useful points; representation.

Worked example

On $]0, +\infty[$, $g(x) = x + \frac{4}{x}$ checks $g'(x) = 1 - \frac{4}{x^2} = \frac{(x-2)(x+2)}{x^2}$. The denominator is positive; g' is negative on $]0, 2[$ then positive on $]2, +\infty[$. The function allows a minimum $g(2) = 4$.

Session 23 — Definition and curve of the reciprocal function

Reciprocal function

15/12/2026 • B : 2 h – C : 1 h

Objectives of the session

· Determine the domain.

· Build the curve.

· Use symmetries.

Fluency · 6 to 8 minutes, without calculator

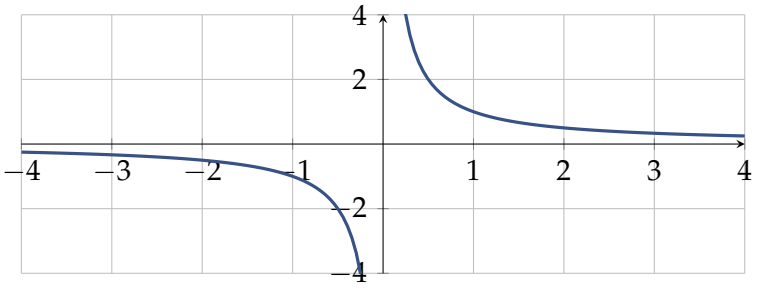
1. Reduce 3 $x + 5 - x + 2$. $2x + 7$
2. To give the prohibited value of $\dfrac{x+1}{x}$. $x = 0$
3. Calculate $\dfrac{1}{5}$ and $-\dfrac{1}{5}$. $\dfrac{1}{5}$ and $-\dfrac{1}{5}$
4. Derive 3 $x^2 - 5x + 1$. $6x - 5$
5. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: determine the domain and build the curve.

Research activity

Calculate $\dfrac{1}{x}$ for $x = -4, -2, -1, -\dfrac{1}{2}, \dfrac{1}{2}$, 1, 2, 4. Place points and describe symmetries.
$x =$ Is 0 absent?



Detailed correction · Professor edition

The values are respectively $-\frac{1}{4}, -\frac{1}{2}, -1, -2, 2, 1, \frac{1}{2}, \frac{1}{4}$. The domain excludes 0; the curve is symmetric to the original. Verbate the data that triggers the choice of method.

Trace written

The reciprocal function is $f(x) = \dfrac{1}{x}$ on $\mathbb{R} \setminus \{0\}$. Its curve is a symmetric hyperbole in relation to originally.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

The values are respectively $-\frac{1}{4}, -\frac{1}{2}, -1, -2, 2, 1, \frac{1}{2}, \frac{1}{4}$. The domain excludes 0; the curve is symmetric with respect to origin.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 5 + \dfrac{7}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{25}{x}$.

Detailed correction · Professor edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 5$. $f'(x) = -\dfrac{7}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{25}{x^2}$ and minimum is reached in $x = 5$ of value 10.

Reference in the manual

Supplements: Calao, courses p. 88·93; exercises p. 94, n° 1-5.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 101, exercise 42

The purpose of this exercise is to draw the curves representative of the functions f and g defined on $\mathbb{R}^*$ by: $f(x) = \dfrac{2}{x}$ and $g(x) = 4 - \dfrac{3}{x}$.

1. Complete the table of values for $x \in \{-5; -4; -2; -0{,}5; -0{,}2; -0{,}1; 0{,}1; 0{,}2; 0{,}5; 1; 2; 4; 5\}$.
2. Using a table of values, construct the curves representative of the functions f and g.

Detailed correction · Professor edition

Calculate each image according to the prohibited value 0. Both curves are asymptote vertical $x = 0$. Horizontal asymptote of f is $y = 0$; that of g is $y = 4$. To control the route, check for example $f(1) = 2$, $f(-1) = -2$, $g(1) = 1$ and $g(-1) = 7$.

Personal work

Resuming Exercise 2 with the following data:
235 and the rate or common difference / ratio by 7. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 24 — Behavior at bounds and asymptotes

Reciprocal function

17/12/2026 • B : 1 h – C : 2 h

Objectives of the session

· Describe the behaviour at the bounds.

· Identify asymptotes.

· Read a representation.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 per cent of 120. 6

2. Calculate $\frac{1}{6}$ and ~~$-\frac{1}{6}$~~. $\frac{1}{6}$ and $-\frac{1}{6}$

3. Derive 3 $x^2 - 5x + 1$. $6x - 5$

4. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$

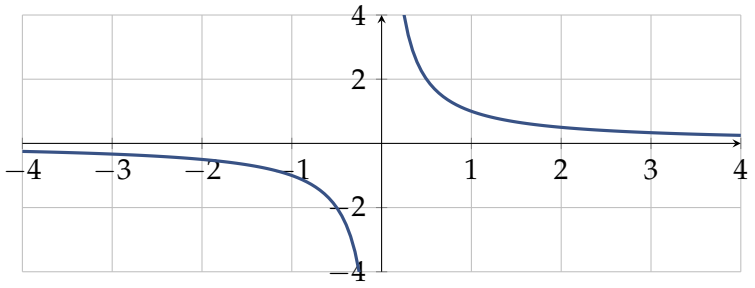
5. To give the prohibited value of $\frac{x+1}{x}$. $x = 0$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, have the keywords reformulated: describe the behaviour at the bounds and identify the Asymptotes.

Research activity

Using a table of values close to 0 and for large values of $|x|$, describe $\frac{1}{x}$. Translate each behavior by an asymptote and a sentence.



Detailed correction · Professor edition

Nearly 0, the values become very large in absolute value: asymptote $x = 0$. When $|x|$ grows, $\frac{1}{x}$ is close to 0: asymptote $y = 0$. Verbate the data that triggers the choice method.

Trace written

For $f(x) = \frac{1}{x}$, $x = 0$ is vertical asymptote and $y = 0$ horizontal asymptote.

approach without cutting them off.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

Nearly 0, the values become very large in absolute value: asymptote $x = 0$. When $|x|$ grows, $\dfrac{1}{x}$ is close to 0: asymptote $y = 0$.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 6 + \dfrac{8}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{36}{x}$.

Detailed correction · Professor edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 6$. $f'(x) = -\dfrac{8}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{36}{x^2}$ and minimum is reached in $x = 6$ of value 12.

Reference in the manual

Supplements: Calao, courses p. 88·93; exercises p. 94, n° 4 to 9.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 95, exercise 16

We want to plot the representative curve of the function g defined on $\mathbb{R}^*$ by: $g(x) = 3 - \dfrac{2}{x}$.

1. Complete an array of values for

$$x \in \{-1\,000; -500; -200; -100; -50; 50; 100; 200; 500; 1\,000\}.$$

   What can be inferred from this for the behaviour of g in $-\infty$ and in $+\infty$?

2. Complete an array of values for $x = -0{,}5; -0{,}1; -0{,}01; 0{,}01; 0{,}1; 0{,}5$. What can be done about deduce for the behaviour of g 0?
3. Copy and complete the table of values in the manual, then plot the representative curve of function g.

Detailed correction · Professor edition

When $x \to \pm\infty$, $-2/x \to 0$, therefore $g(x) \to 3$. When $x \to 0^-$, $-2/x \to +\infty$; when $x \to 0^+$, $-2/x \to -\infty$. Asymptotes are $x = 0$ and $y = 3$.

Personal work

Resuming Exercise 3 with the following data:
240 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 25 — Derivative and direction of variation

Reciprocal function

05/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Derivate the reciprocal function.

· Justify variations.

· Draw a picture.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient of a 10 per cent increase. $1+\frac{10}{100}=1,10$
2. Derive 3 x^2-5x+1. $6x-5$
3. Solve $(x-2)(x+1)=0$. $x=-1$ **or** $x=2$
4. To give the prohibited value of $\frac{x+1}{x}$. $x=0$
5. Calculate $\frac{1}{7}$ and ~~$-\frac{1}{7}$~~. $\frac{1}{7}$ and $-\frac{1}{7}$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: drift the reciprocal function and justify the variations.

Research activity

Based on the rate of increase of $f(x)=\frac{1}{x}$, find $f'(x)=-\frac{1}{x^2}$. Study his sign and check consistency with the graph.

Detailed correction · Professor edition

The square x^2 is positive for $x\neq 0$, so the derivative is strictly negative on each of the intervals
Inform the data that triggers the choice of method.

Trace written

For $x\neq 0$, $f'(x)=-\frac{1}{x^2}<0$. The reciprocal function decreases at each of the intervals $]-\infty,0[$ and $]0,+\infty[$.

Worked example

Retrieve activity data and write the solution in order: definition of objects,
property used, exact calculation or approach, then response phrase. The model response figure
in the teaching edition.

Detailed correction · Professor edition

The square x^2 is positive for $x \neq 0$, so the derivative is strictly negative on each of the intervals of the field.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 2 + \dfrac{4}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{4}{x}$.

Detailed correction · Professor edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 2$. $f'(x) = -\dfrac{4}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{4}{x^2}$ and minimum is reached in $x = 2$ of value 4.

Reference in the manual

Supplements: Calao, courses p. 89·93; exercises p. 94, n° 1 to 3, and p. 96, n° 10 to 13. reproduced in this session is identified below by its page and number; the reference allows also find the figures and digital resources of the manual.

Manuel Calao · p. 94, exercise 3

Calculate the derivatives of the following functions defined on $\mathbb{R}^*$.

a. $f(x) = 2x - 3 + \dfrac{4}{x}$

b. $g(x) = 3x - 5 - \dfrac{1}{x}$

c. $f(x) = 7x^3 - 2x + \dfrac{1}{x}$

d. $g(x) = 3x^2 - 5x - 2 + \dfrac{4}{x}$

Detailed correction · Professor edition

Derivatives are: $f'(x) = 2 - 4/x^2$; $g'(x) = 3 + 1/x^2$; $f'(x) = 21x^2 - 2 - 1/x^2$; $g'(x) = 6x - 5 - 4/x^2$.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 26 — Combinations with polynomial functions

Reciprocal function

07/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Derive a combination.

· Reduce expression.

· Organize a study.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x+8-x+2$. $2x+10$
2. Resolve $(x-2)(x+1)=0$. $x=-1$ **or** $x=2$
3. To give the prohibited value of $\dfrac{x+1}{x}$. $x=0$
4. Calculate $\dfrac{1}{8}$ and ~~$\dfrac{1}{8}$~~. $\dfrac{1}{8}$ and $-\dfrac{1}{8}$
5. Derive 3 x^2-5x+1. $6x-5$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: derive a combination and reduce an expression.

Research activity

A company produces $x>0$ hundreds of objects and its average cost is $C(x)=2x+30+\dfrac{200}{x}$. Calculate $C'(x)$, factoring its numerator and seeking the level of production minimizing the cost Medium.

Detailed correction · Professor edition

$C'(x)=2-\dfrac{200}{x^2}=\dfrac{2(x-10)(x+10)}{x^2}$. The minimum is reached for $x=10$, or 1,000 objects. verbalize the data that triggers the choice of method.

Trace written

To Study $P(x)+\dfrac{k}{x}$, determine the domain, calculate the derivative, reduce to the same denominator, study his sign and then draw up variations.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$C'(x) = 2 - \frac{200}{x^2} = \frac{2(x-10)(x+10)}{x^2}$. The minimum is reached for $x = 10$, or 1,000 objects.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 3 + \frac{5}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \frac{9}{x}$.

Detailed correction · Professor edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 3$. $f'(x) = -\frac{5}{x^2} < 0$. For g, $g'(x) = 1 - \frac{9}{x^2}$ and minimum is reached in $x = 3$, value 6.

Reference in the manual

Supplements: Calao, p. 96, No. 14-21. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 96, exercise 20

Let f the function defined on $\mathbb{R}^*$ by $f(x) = 4x + 1 + \frac{9}{x}$.

1. Show that for all real x Non-zero: $f'(x) = \frac{(2x-3)(2x+3)}{x^2}$.
2. For all real x non-zero, study the sign of $f'(x)$.
3. Deduce changes in function f.

Detailed correction · Professor edition

$f'(x) = 4 - 9/x^2 = (4x^2 - 9)/x^2 = ((2x-3)(2x+3))/x^2$. The denominator is positive: f' is positive on $]-\infty; -3/2[\cup]3/2; +\infty[$ and negative $]-3/2; 0[\cup]0; 3/2[$. The variations are deduced on each of the intervals separated by $-3/2$, 0 and 3/2.

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 27 — Signing and factoring

Reciprocal function

12/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Factorize.

· Build a signboard.

· Respect the prohibited values.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 180. 36

2. To give the prohibited value of $\dfrac{x+1}{x}$. $x = 0$

3. Calculate $\dfrac{1}{9}$ and ~~-1~~$\dfrac{1}{9}$. $\dfrac{1}{9}$ and $-\dfrac{1}{9}$

4. Derive 3 $x^2 - 5x + 1$. $6x - 5$

5. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: factor and construct a signboard.

Research activity

Study the sign of $\dfrac{(x-2)(x+3)}{x}$. Build the table with the three critical values, distinguish zero and forbidden value, then solve the inequitable $\dfrac{(x-2)(x+3)}{x} \geq 0$.

Detailed correction · Professor edition

The critical values are -3, 0, 2. The solution is $[-3, 0[\cup [2, +\infty[$. Verbate the data that triggers the choice of method.

Trace written

In a quotient, the sign depends on the numerator and denominator.
excluded from the table, even if a factor is formally simplified.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

The critical values are -3, 0, 2. The solution is $[-3, 0[\cup [2, +\infty[$.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 4 + \frac{6}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \frac{16}{x}$.

Detailed correction · Professor edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 4$. $f'(x) = -\frac{6}{x^2} < 0$. For g, $g'(x) = 1 - \frac{16}{x^2}$ and minimum is reached in $x = 4$, value 8.

Reference in the manual

Supplements: Calao, pp. 96-101. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 96, exercise 21

Let g the function defined on $\mathbb{R}^*$ by $g(x) = x + 10 - \frac{1}{x}$.

1. Show that for all real x Non-zero: $g'(x) = \frac{x^2 + 1}{x^2}$.
2. For all real x non-zero, study the sign of $g'(x)$.
3. Deduce changes in function g.

Detailed correction · Professor edition

$g'(x) = 1 + 1/x^2 = (x^2 + 1)/x^2 > 0$ for all $x \neq 0$. The function is therefore strictly increasing on each interval $]-\infty; 0[$ and $]0; +\infty[$.

Personal work

Resuming Exercise 2 with the following data:
255 and rate or common difference / ratio by 5. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.

Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 28 — Derivatives of polynomials of not more than 3 degrees

Reciprocal function

14/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Drain a polynomial.

· Study a derivative.

· Reduce variations.

Fluency · 6 to 8 minutes, without calculator

1. Write the multiplier coefficient with an increase of 25 per cent. $1 + \frac{25}{100} = 1,25$
2. Calculate $\frac{1}{3}$ and ~~$-\frac{1}{3}$~~. $\frac{1}{3}$ and $-\frac{1}{3}$
3. Derive 3 $x^2 - 5x + 1$. $6x - 5$
4. Solve $(x-2)(x+1) = 0$. $x = -1$ **or** $x = 2$
5. To give the prohibited value of $\frac{x+1}{x}$. $x = 0$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: drift a polynomial and study a derivative.

Research activity

For $f(x) = 2x^3 - 3x^2 - 12x + 5$, calculate $f'(x)$, factoring, studying his sign and deducing from it the changes in f.

Detailed correction · Professor edition

$f'(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1)$. The function grows on $]-\infty, -1]$, decreases on $[-1, 2]$then Make the data that triggers the choice of method become worded.

Trace written

$(ax^3 + bx^2 + cx + d)' = 3ax^2 + 2bx + c$. The derivative sign gives the direction of variation of the function.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$f'(x) = 6x^2 - 6x - 12 = 6(x-2)(x+1)$. The function grows on $]-\infty, -1]$, decreases on $[-1, 2]$then growth.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 5 + \frac{7}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \frac{25}{x}$.

Detailed correction · Professor edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 5$. $f'(x) = -\frac{7}{x^2} < 0$. For g, $g'(x) = 1 - \frac{25}{x^2}$ and minimum is reached in $x = 5$ of value 10.

Reference in the manual

Supplements: Calao, pp. 94-96. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 94, exercise 13

Let f the function defined on $\mathbb{R}^*$ by: $f(x) = \frac{1}{x}$.

1. Calculate $f'(x)$.
2. Show that for all real x Non-zero: $f'(x) = \frac{-1}{x^2}$.
3. After studying the sign of $f'(x)$, to draw up the table of variations of f.

Detailed correction · Professor edition

$f'(x) = -1/x^2 < 0$ for all $x \neq 0$. The reciprocal function is therefore strictly decreasing on each interval $]-\infty; 0[$ and $]0; +\infty[$. These two intervals should not be combined into one Decreasing range.

Personal work

Resuming Exercise 3 with the following data:
260 and the rate or common difference / ratio by 6. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.

Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 29 — Comprehensive study and representation

Reciprocal function

19/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Conduct a comprehensive study.

· Draw a coherent curve.

· Interpret an extreme.

Fluency · 6 to 8 minutes, without calculator

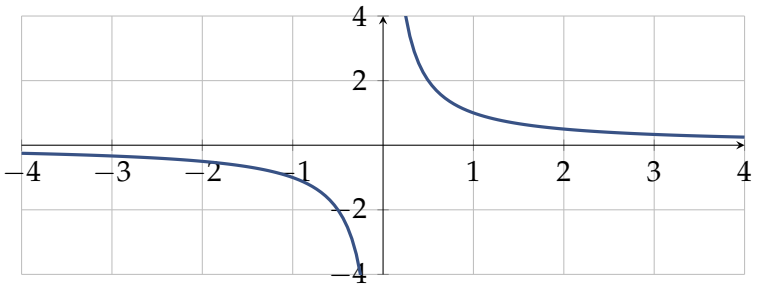
1. Reduce 3 $x+4-x+2$. $2x+6$
2. Derive 3 x^2-5x+1. $6x-5$
3. Solve $(x-2)(x+1)=0$. $x=-1$ **or** $x=2$
4. To give the prohibited value of $\dfrac{x+1}{x}$. $x=0$
5. Calculate $\dfrac{1}{4}$ and ~~$-\dfrac{1}{4}$~~. $\dfrac{1}{4}$ and $-\dfrac{1}{4}$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: conduct a complete study and plot a curve coherent.

Research activity

Studying completely $g(x)=x+\dfrac{9}{x}$ on $]0,+\infty[$: domain, derivative, sign, variations, minimum, behaviour at the bounds and outline.



Detailed correction · Professor edition

$g'(x)=\dfrac{(x-3)(x+3)}{x^2}$; minimum $g(3)=6$; $g(x)$ tend towards $+\infty$ in 0^+ and in $+\infty$. Verbating the data that triggers the choice of method.

Trace written

A complete study includes: domain, derivative, sign, variations, limitations or behaviour in the bounds, remarkable points and consistent graphic.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$g'(x) = \dfrac{(x-3)(x+3)}{x^2}$; minimum $g(3) = 6$; $g(x)$ tend towards $+\infty$ in 0^+ and in $+\infty$.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 6 + \dfrac{8}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{36}{x}$.

Detailed correction · Professor edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 6$. $f'(x) = -\dfrac{8}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{36}{x^2}$ and minimum is reached in $x = 6$ of value 12.

Reference in the manual

Supplements: Calao, pp. 98-103. The exercise reproduced in this session is identified below. by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 98, financial year 31

A company specialising in industrial ovens can manufacture between 20 and 80 depending on the years. x the number of ovens manufactured annually.
The graphic representation given in the manual is that of a function C defined on L-range $[20;80]$. For all integer values of x, $C(x)$ is the unit cost of production Average (in €).

1. What is the average unit production cost when 25 ovens are produced and then when 70 Ovens are produced?
2. Which productions correspond to an average unit cost of 20,000 € ?
3. What is the minimum unit cost of production? What production does it correspond to?

Detailed correction · Professor edition

The numerical responses are read on the curve of p. 98. Require reading traits and units.
The graphic minimum must be identified by the order of the lowest point and then linked to its abscissa; a graphic reading remains close.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 30 — Assessment and evaluation

Reciprocal function

21/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Mobilize domain, derivative and sign.

· Write a study.

· Correct methodically.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5% of 100. 5
2. Resolve $(x-2)(x+1)=0$. $x=-1$ **or** $x=2$
3. To give the prohibited value of $\dfrac{x+1}{x}$. $x=0$
4. Calculate $\dfrac{1}{5}$ and ~~$\dfrac{1}{5}$~~. $\dfrac{1}{5}$ and $-\dfrac{1}{5}$
5. Derive 3 x^2-5x+1. $6x-5$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, have the keywords reformulate: mobilize domain, derivative and sign and write a study.

Research activity

Correcting a deliberately wrong study of $h(x)=x+\dfrac{4}{x}$: area announced $\mathbb{R}$, derivative $1+\dfrac{4}{x^2}$, Tableau sans rupture en 0. Rewrite a complete solution.

Detailed correction · Professor edition

The domain is $\mathbb{R}\setminus\{0\}$ and $h'(x)=1-\dfrac{4}{x^2}$. The table is separated into 0 and takes account of -2 and 2. Verbate the data that triggers the choice of method.

Trace written

Control the four objects separately: domain, derivative expression, derivative sign, Variations. A one-step error should not mask the following.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

The domain is $\mathbb{R} \setminus \{0\}$ and $h'(x) = 1 - \dfrac{4}{x^2}$. The table is separated into 0 and takes account of -2 and 2.

Training · essential

1. Provide the domain and asymptotes of $f(x) = 2 + \dfrac{4}{x}$.
2. Calculate $f'(x)$ and study his sign.
3. Study on $]0, +\infty[$ function $g(x) = x + \dfrac{4}{x}$.

Detailed correction · Professor edition

The domain is $\mathbb{R} \setminus \{0\}$; asymptotes $x = 0$ and $y = 2$. $f'(x) = -\dfrac{4}{x^2} < 0$. For g, $g'(x) = 1 - \dfrac{4}{x^2}$ and minimum is reached in $x = 2$ of value 4.

Reference in the manual

Supplements: Calao, p. 100 to 103. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 98, exercise 32

Every week, a liquid detergent company estimates that the cost of production (in euros) can be modelled by a function C given by: $C(x) = x^2 + 60x + 121$, where x is the volume of detergent (m) ^{3}), $x \in [1;30]$.
Let f the function representing the average cost of production per m ^{3} liquid detergent.

1. Show that for everything $x \in [1;30]$, $f(x) = x + 60 + \dfrac{121}{x}$.
2. Calculate $f'(x)$. Show that for everything $x \in [1;30]$, $f'(x) = \dfrac{(x-11)(x+11)}{x^2}$.
3. Draw up the table of changes in f.
4. What is the minimum average cost of production? For what amount of detergent is it reached ?

Detailed correction · Professor edition

$f(x) = C(x)/x = x + 60 + 121/x$. Then $f'(x) = 1 - 121/x^2 = ((x-11)(x+11))/x^2$. On $[1;30]$, f' is negative up to 11, then positive. The minimum is reached for $x = 11$ and is $f(11) = 82$ € by m^3.

Personal work

Resuming Exercise 3 with the following data:
270 and the rate or common difference / ratio by 2. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
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